Worksheet

Infinite Series

Determine whether each of the infinite series converges or diverges. If the infinite series converges, write, “the infinite series converges”. If the infinite series diverges, write “the infinite series diverges”.

1. $\displaystyle\sum_{k=1}^{\infty}\sqrt{\frac{2}{k}}$

2. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt{k(k+1)}}$

3. $\displaystyle\sum_{k=1}^{\infty}\frac{(k+10)^2}{k^{10/3}}$

4. $\displaystyle\sum_{k=1}^{\infty}\frac{5}{4^k+3}$

5. $\displaystyle\sum_{k=1}^{\infty}\frac{\sqrt{k}}{5^k}$

6. $\displaystyle\sum_{k=1}^{\infty}\frac{\sqrt[4]{k}+1}{\sqrt{k^3+1}}$

7. $\displaystyle\sum_{k=1}^{\infty}\frac{\ln k}{k^2}$

8. $\displaystyle\sum_{k=1}^{\infty}\frac{3k^2+6}{4k^2+12k+8}$

9. $\displaystyle\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\frac{1}{7\cdot 9}+\frac{1}{9\cdot 11}+\cdots+\frac{1}{(2k-1)(2k+1)}+\cdots$